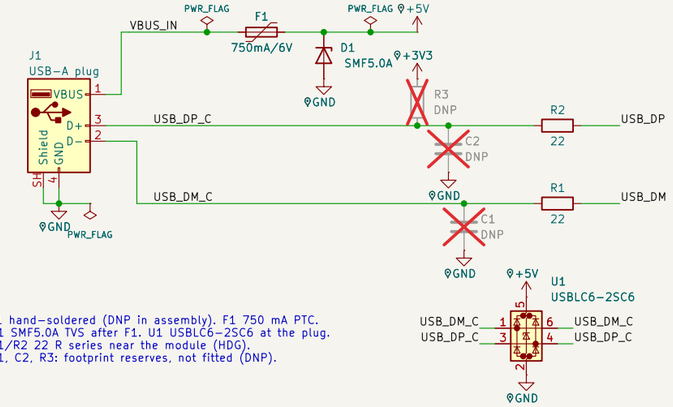
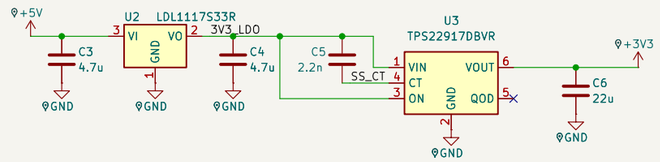
USB INPUT & PROTECTION



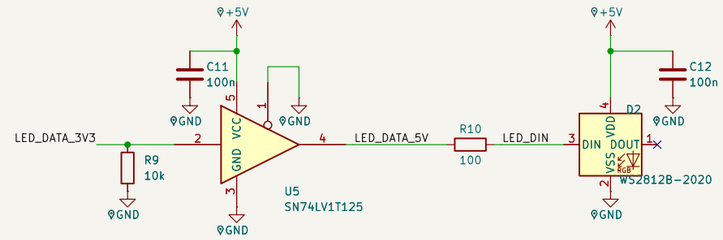
J1 hand-soldered (DNP in assembly). F1 750 mA PTC.
D1 SMF5.0A TVS after F1. U1 USBLC6-2SC6 at the plug.
R1/R2 22 R series near the module (HDG).
C1, C2, R3: footprint reserves, not fitted (DNP).

3.3 V SUPPLY + SOFT-START



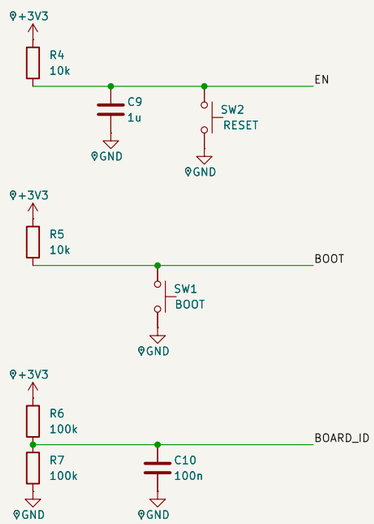
U2 LDL1117 5 V -> 3.3 V: P = 1.7 V x 0.25 A avg = 0.43 W; tab copper + 4 vias.
U3 TPS22917 soft-start: CT (C5) to VIN per TI DS pin table;
tR = 1.6 us/pF x 2.2 nF = 3.5 ms (at 3.6 V).
Inrush = C x dV/dt = (22u + 10u + 0.1u + module) x 3.3 V / 3.5 ms = ~30 mA + module.
QOD open: no quick output discharge on the main rail.

STATUS LED + 3.3 V -> 5 V LEVEL SHIFTER



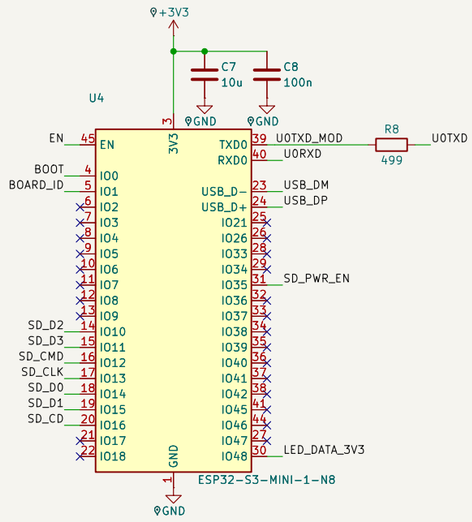
U5 SN74LV1T125 (74LVC1G125 symbol, same pinout): single-supply
translator at VCC 5 V. VIH 2.03 V max -> 3.3 V input is valid. OE tied low.
R9 holds DIN low during boot (LED dark). R10 100 R at DIN.

RESET / BOOT / BOARD ID



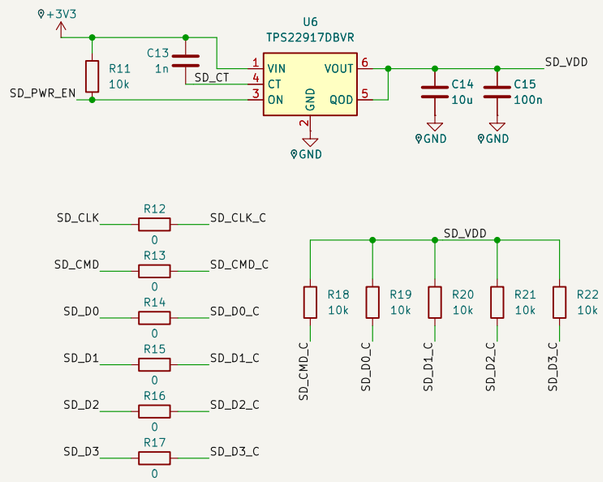
EN RC: R4 10k / C9 1u (module DS). SW2 = reset.
BOOT (IO0): R5 10k pull-up, no cap. Hold SW1 at reset -> download.
Board ID (IO1, ADC): RX R7 = 100k -> 1.65 V;
TX R7 = 22k (C25768) -> 0.60 V. TX firmware ignores BOARD_ID.

ESP32-S3 MODULE



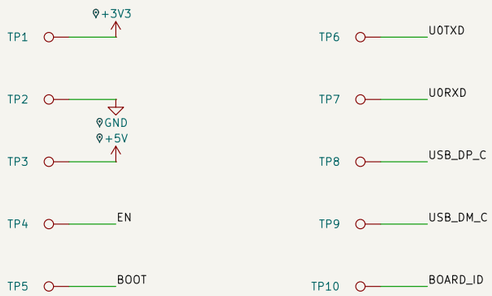
U4 = ESP32-S3-MINI-1-N8 only (IO35 used; no PSRAM variants).
Native USB on IO19/IO20 (USB_D-/D+). C7 10u + C8 100n at the 3V3 pin.
SDMMC via GPIO matrix: pin order follows the layout.

microSD + POWER SWITCH (RX ONLY)



RX only: every part in this block is Fit=RX (not fitted on TX).
U6 TPS22917: CT (C13 1 nF) to VIN, tR = ~1.6 ms. QOD tied to VOUT (fast discharge).
R11 pulls ON high: SD power on by default; IO35 can switch it off.
R12-R17 0 R series (tune in layout): R18-R22 pull-ups to the switched SD_VDD.
J3 SHOU HAN TF PUSH (push-push). Card detect switches to GND on insert [polarity unverified].

TEST PADS (bottom side, pogo jig)



GENERAL NOTES

Fit field: ALL = TX + RX builds; RX = receiver only (DNP on TX); DNP = footprint only (X).
TX build: microSD block not fitted, R7 = 22k. J1 hand-soldered in every build.
Inter-block nets use local labels; power symbols are sheet-local (net names /GND, /+3V3, /+5V).
Test pads: bottom-side copper pads for the production pogo jig (not in BOM/CPL).
This sheet was laid out by tools/gen_sch.py once (2026-11-10);
filexfer-pcb.kicad_sch is now the master. Edit it in KiCad.
Design notes: docs/stage2-parts.md, docs/stage4-notes.md, docs/schematic-style.md.

One board for TX and RX builds; Fit field selects the build
ESP32-S3-MINI-1-N8, USB-A plug, microSD (RX build)
Sheet 1/1: Main (USB input, 3.3 V supply, module, microSD, LED, buttons, test pads)
Author: Richmond Osei-Baah

FileXfer

Sheet: /
File: filexfer-pcb.kicad_sch

Title: FileXfer rev A

Size: A3	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.0.4		Id: 1/1